

Solving Equations

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Learning Goals

- Understand what a simultaneous equation represents
- Solving systems with one variable
- Solve systems with substitution and elimination
- Check if your solution makes sense

1 Why Equations?

You might be thinking about x and y . These are simply just a *place holder* for a number you don't know yet. We call it a **variable**.

Example 1.1 I have a number, such that when I multiply it by 4 and then add 2, I get 18. What is my number?

It's very easy to see that my number is 4. But we will put it in equation terms:

$$4x + 2 = 18$$

I'm just using x as a place holder for the number that I don't know yet.

Example 1.2 I have two numbers. They add to 10, and three times the first number take away the second number is 2. What are the two numbers?

Now we have two numbers, we can use x and y .

$$\begin{aligned}x + y &= 10 \\ 3x - y &= 2\end{aligned}$$

Problem 1.1 Sarah has some apples and oranges. She has 3 more apples than oranges. If she has 15 pieces of fruit altogether, how many apples and oranges does she have?

Problem 1.2 A movie ticket costs \$8 and popcorn costs \$5. If Marcus spent \$46 on 7 items total (tickets and popcorns), how many of each did he buy?

2 One Variable — The Basics

You need to treat each equation like a balance. Whatever you do to one side, you **must** do it to the other side too, so the balance don't tip.

We can look at the previous example:

$$4x + 2 - 2 = 18 - 2 \implies 4x = 16 \implies x = 4$$

Kind of similar to working backwards, right?

Problem 2.1 Solve for x : $2x - 5 = 11$

Problem 2.2 Solve for x : $\frac{x}{3} + 4 = 7$

Problem 2.3 Solve for x : $5(x - 2) = 30$

Problem 2.4 Solve for x : $7x + 3 = 2x + 18$

3 Two Variables

Lets look at our example from before.

$$x + y = 10$$

$$3x - y = 2$$

3.1 Substitution

We are going to *isolate* the variables.

$$x + y = 10 \implies y = 10 - x$$

and

$$3x - y = 2 \implies 3x = 2 + y \implies 3x - 2 = y \implies y = 3x - 2$$

But wait, we now have y = something in two ways. Since both are representing the same thing, lets put them together:

$$10 - x = 3x - 2$$

We can solve this now for x , and we get $x = 3$. WAIT! We aren't done yet. We only found x , but it's very important you also find y .

To do that, we can just change x into 3 in one of the equations we had.

$$3 + y = 10 \implies y = 7$$

3.2 Elimination

Lets take the same equations again:

$$\begin{aligned}x + y &= 10 \\ 3x - y &= 2\end{aligned}$$

In the end, the thing on the left side $x + y$ is just another number, and we know that number is 10. Same thing with $3x - y$. So actually we can just add both of these equations together.

$$\begin{aligned}x + y + (3x - y) &= 10 + 2 \\ 4x &= 12 \\ x &= 3\end{aligned}$$

Same thing as before, now we have x , we substitute it back and get $y = 7$.

3.3 Presenting & Checking Your Solution

After you've solved the equation and you have your x and y , write it like this: $(x, y) = (3, 7)$. To check your solution, you need to substitute your answers back.

$$\begin{aligned}3 + 7 &= 10 \\ 3 \times 3 - 7 &= 2\end{aligned}$$

Which all works out!

3.4 Practice Problems

Problem 3.1 Solve the system using substitution or elimination:

$$\begin{aligned}x + 2y &= 7 \\ 2x - y &= 4\end{aligned}$$

Problem 3.2 Solve the system using substitution or elimination:

$$\begin{aligned}3x + y &= 11 \\ x - y &= -3\end{aligned}$$

Problem 3.3 Solve the system using substitution or elimination:

$$\begin{aligned}2x + 3y &= 12 \\ 4x - y &= 5\end{aligned}$$

4 Word Problems

Problem 4.1 A farmer has a total of 50 animals, consisting of chickens and cows. If there are 20 more chickens than cows, how many of each type of animal does the farmer have?

Problem 4.2 A bookstore sold a total of 120 books in one day. If the number of fiction books sold was twice the number of non-fiction books sold, how many of each type were sold?

Problem 4.3 A class has a total of 30 students. If there are 5 more girls than boys in the class, how many boys and girls are there?

Problem 4.4 A train leaves a station traveling at a speed of 60 km/h. Another train leaves the same station 30 minutes later, traveling at a speed of 90 km/h. How far from the station will the two trains meet?

Problem 4.5 A rectangle has a length that is twice its width. If the perimeter of the rectangle is 48 cm, what are the dimensions of the rectangle?

Problem 4.6 The sum of three consecutive integers is 72. What are the integers?

Problem 4.7 A car rental company charges a flat fee of \$50 plus \$0.20 per mile driven. If a customer pays \$70, how many miles did they drive?

Problem 4.8 A farmer has a total of 35 heads of chickens and rabbits. If the total number of feet is 94, how many chickens and how many rabbits does the farmer have?